Aashray Jain

M.Tech Candidate in Artificial Intelligence & Machine Learning

NIMS Marik Institute of Computing, AI, Robotics & Cybernetics, NIMS University Rajasthan, Jaipur, India

+91-9201876340 | aashrayjain99@gmail.com | github.com/aashrayjain |
linkedin.com/in/aashray-jain-b26168119 | aashrayjain.github.io

---

To: Selection Committee, CISPA Summer Research (CSR) 2027 Program
Institution: CISPA Helmholtz Center for Information Security, Saarbrücken, Germany
Target Tracks: B5 (Machine Learning, AI & Data Science) & **B3** (Formal Methods, Algorithms & Foundations)
Subject: Letter of Motivation — Visiting Research Internship (Summer 2027)

Dear Members of the Selection Committee,

I am writing to express my strong motivation to join the **CISPA Summer Research (CSR) 2027 Program**. Currently in the first year of my **M.Tech in Artificial Intelligence & Machine Learning** at NIMS University Rajasthan (holding a B.Tech in Electronics & Communication Engineering with an **8.74 / 10.0 CGPA Distinction**), I have spent over four years combining deep learning research with large-scale distributed systems engineering. My objective at CISPA is to investigate the intersection of **Trustworthy & Explainable AI (XAI)** and **Provable Safety Verification in Autonomous Agentic Architectures**, contributing directly to faculty research within Track B5 and Track B3.

Research Interests & Scientific Inquiries:
Contemporary deep learning systems deployed in mission-critical environments face twin fundamental challenges: an acute vulnerability to distribution skew and uncalibrated hallucinations, and an absence of formal verification guarantees when foundation models interact with external environments. My research interests center on two fundamental inquiries:

1. **Robust Representation Learning & Post-Hoc Interpretability:** Designing mathematically grounded loss dynamics and attention mechanisms that maintain diagnostic fidelity under severe empirical class imbalance, coupled with neuroanatomical attribution (e.g., Grad-CAM) to establish verifiable decision boundaries.
2. **Formal Verification & Safety Bounds for Agentic Systems:** Grounding autonomous foundation models in open protocols (Model Context Protocol - MCP) through pre-execution schema verification and transactional sandboxes that provably prevent unauthorized or non-deterministic state transitions.

How My Academic & Research Background Relates to CISPA:
My academic coursework in Advanced Machine Learning, Probabilistic Inference, and Computational Complexity is directly reinforced by empirical research and production-grade engineering:

- **Deep Learning & XAI Research (ArihantQNet):** On a benchmark of 6,400 T1-weighted axial brain MRI scans, I investigated multi-stage Alzheimer's disease stratification under a severe 49.78:1 class skew. To solve this, I derived a *Normalized Square-Root Inverse-Frequency Weighted Cross-Entropy Loss* with AdamW optimization, mathematically compressing the empirical skew to 7.06:1 and attaining 100% sensitivity on rare pathologies. I formulated the novel tri-branch hybrid network **ArihantQNet**—fusing macro-contextual, fine-boundary, and channel-calibrated representations via Quantum-Inspired Multi-Head Cross-Attention (Q-MCA)—and integrated Grad-CAM heatmaps to clinically validate bilateral hippocampal atrophy.
- **Agentic Systems & Safety Verification (Novafy OU):** As a key contributor to the ground-up rebuild of NovaAgent v2 in Python/FastAPI (190+ reviewed pull requests), I architected tools over the **Model Context Protocol (MCP)** and designed a *Human-in-the-Loop formal safety verification layer* for enterprise ERP (MS Dynamics 365 BC). By enforcing pre-execution live metadata validation and non-hallucinatory value substitution, the system guarantees zero schema violations. Furthermore, I formulated provider-side prompt caching protocols achieving ∼99% cached tokens and designed an automated evaluation harness spanning 70+ deterministic scenarios to quantify reasoning drift.
- **Distributed Systems Foundations (TCS & ArihantQ):** Over 3.5 years at Tata Consultancy Services, I engineered high-throughput streaming architectures and real-time risk platforms for Canadian BFSI serving 500K+ users, while collaborating with TCS Rapid Labs on applied AI/ML initiatives. Through my independent lab ArihantQ, I won the Lyzr AI Architect Challenge for algorithmic LLM cost optimization and earned public commendation from the MariaDB Foundation for open-source database innovation.

Why CISPA & Long-Term Vision:
CISPA Helmholtz Center is world-renowned for defining the theoretical foundations and practical frontiers of cybersecurity and trustworthy information processing. Working under CISPA's leading faculty in Machine Learning and Formal Methods would provide me with rigorous mentorship and computational resources to translate empirical agentic architectures into mathematically verifiable, trustworthy systems. This summer research experience will serve as the cornerstone of my Master's thesis and prepare me directly for competitive applications to a fully-funded PhD program in Germany.

I welcome the opportunity to contribute my dual strengths in mathematical formulation and rigorous systems engineering to CISPA's research mission, and I thank you for considering my application.

Sincerely,

Aashray Jain
M.Tech Candidate in Artificial Intelligence & Machine Learning
NIMS Marik Institute of Computing, AI, Robotics & Cybernetics, NIMS University Rajasthan, Jaipur, India